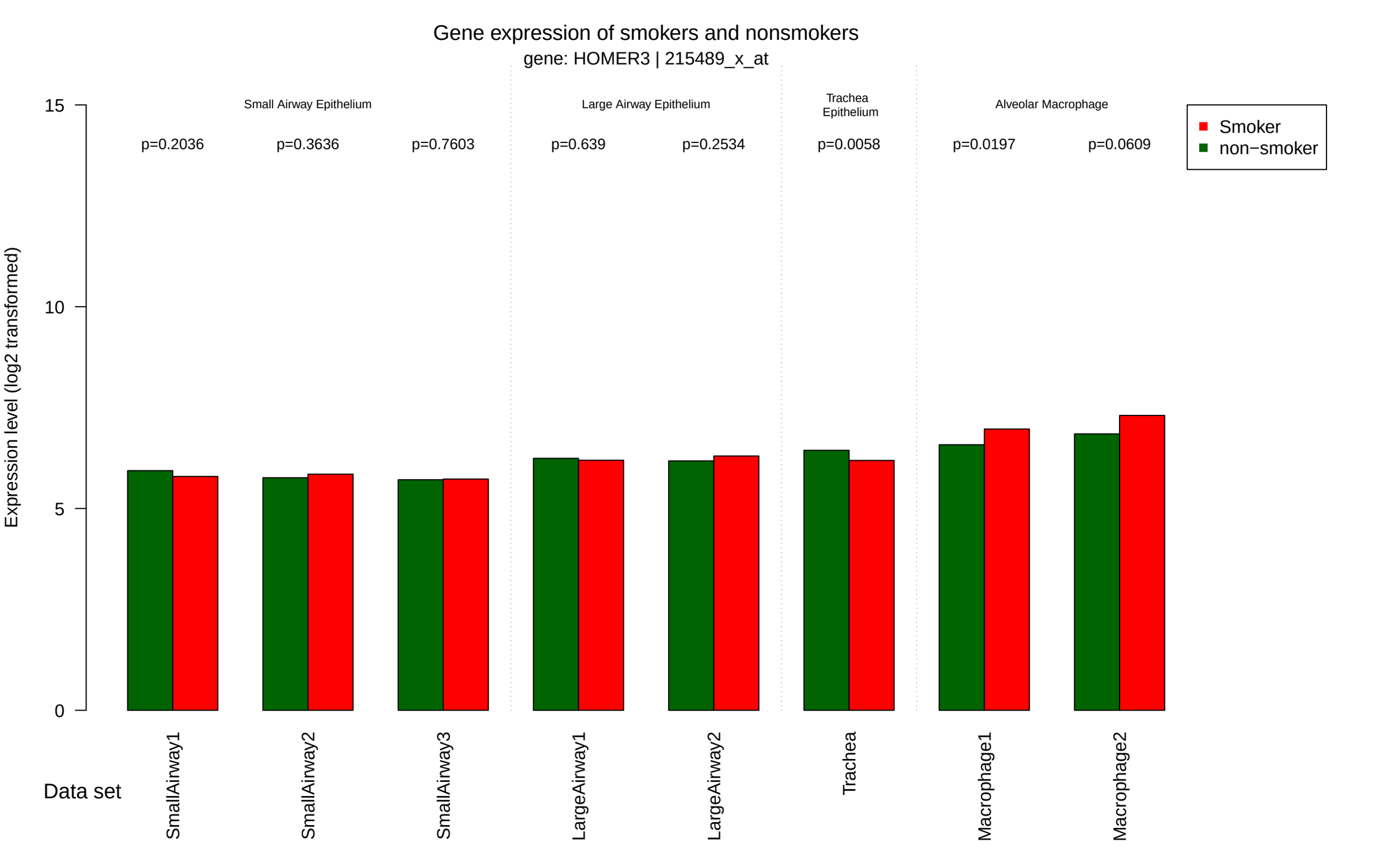
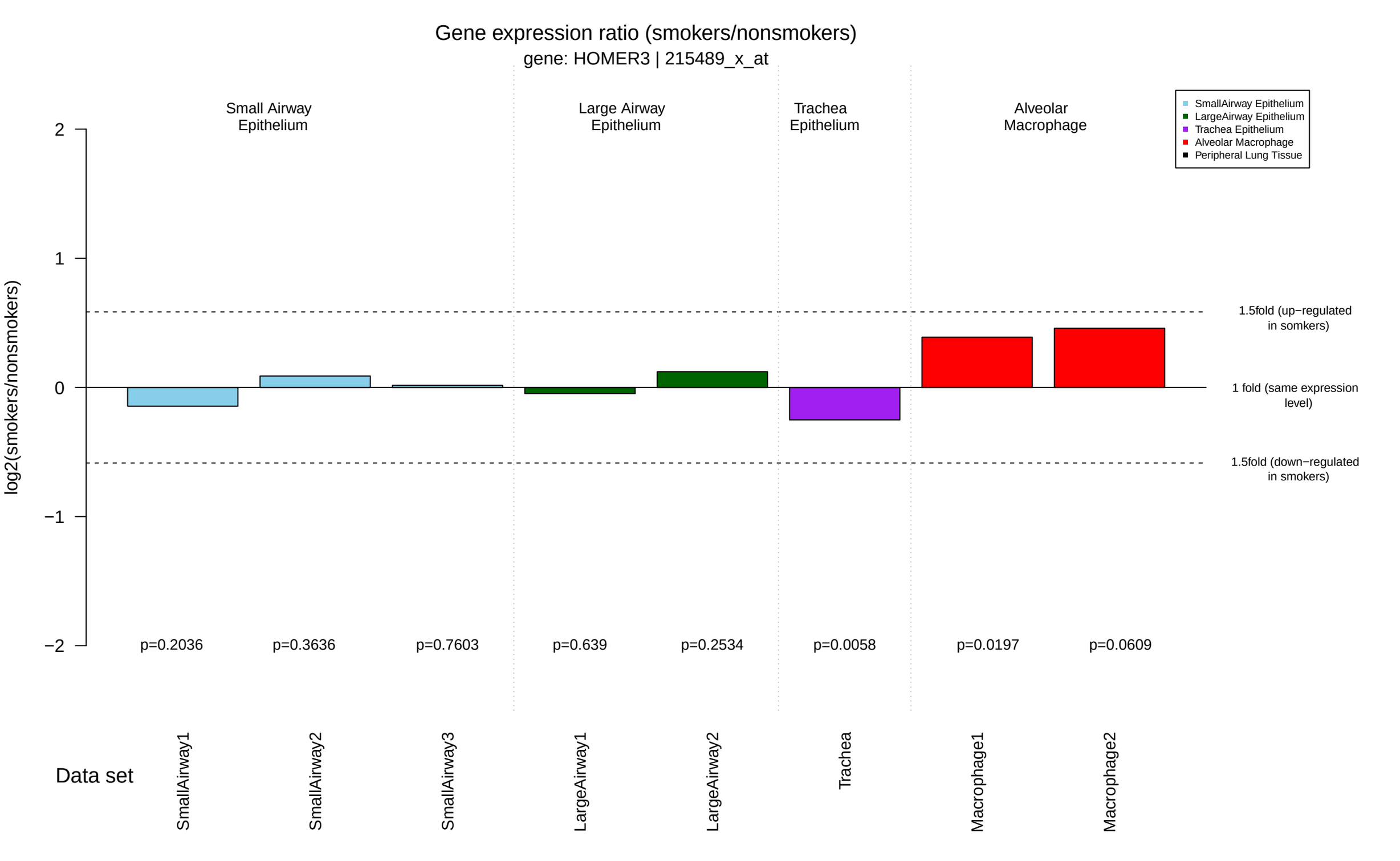
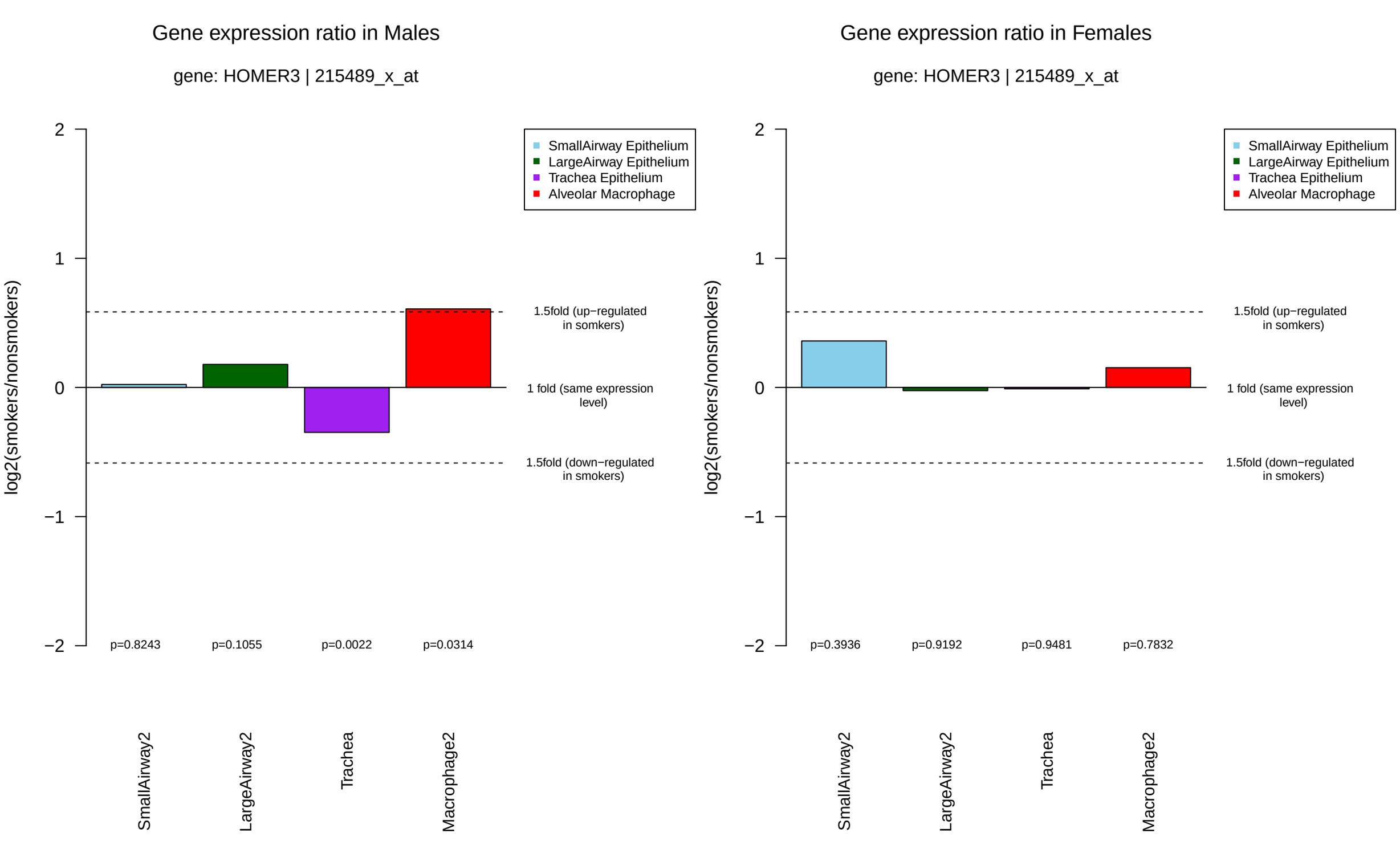
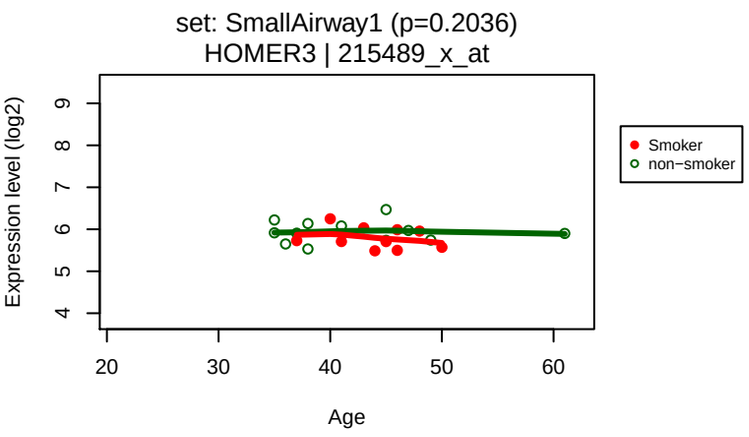
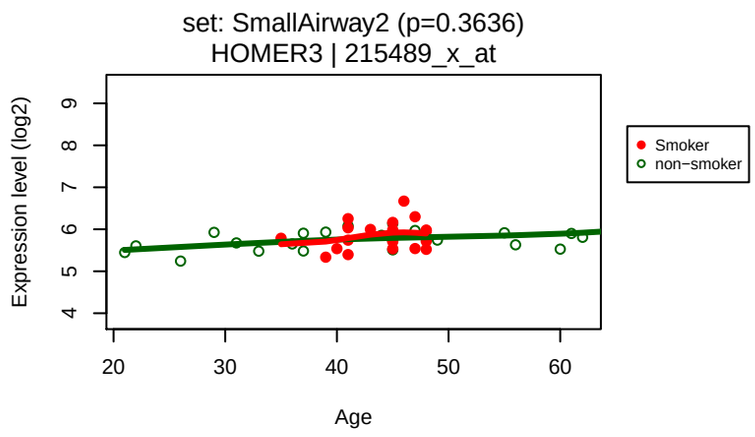
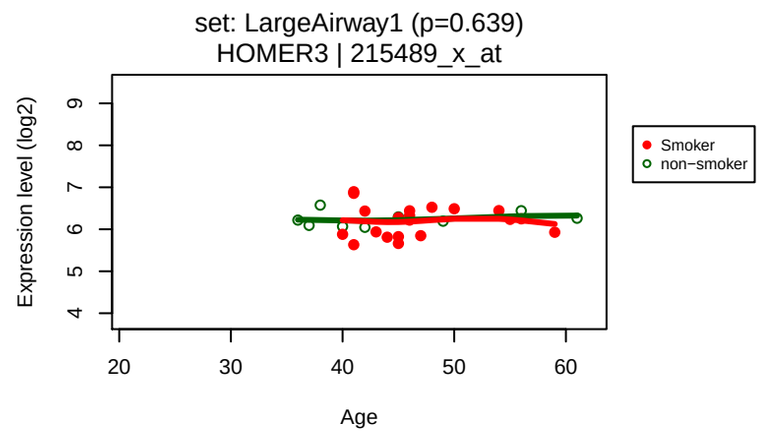

Gene expression of smokers and nonsmokers
gene: HOMER3 | 215489_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2036
p=0.3636
p=0.7603
p=0.639
p=0.2534
p=0.0058
p=0.0197
p=0.0609
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

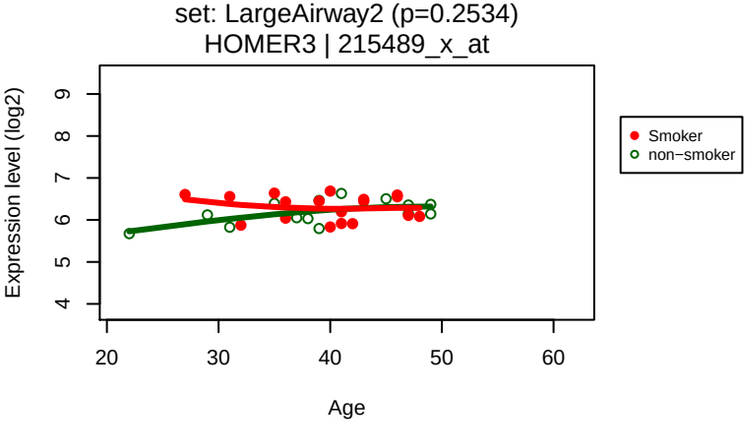

Gene expression ratio (smokers/nonsmokers)
gene: HOMER3 | 215489_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2036
p=0.3636
p=0.7603
p=0.639
p=0.2534
p=0.0058
p=0.0197
p=0.0609
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

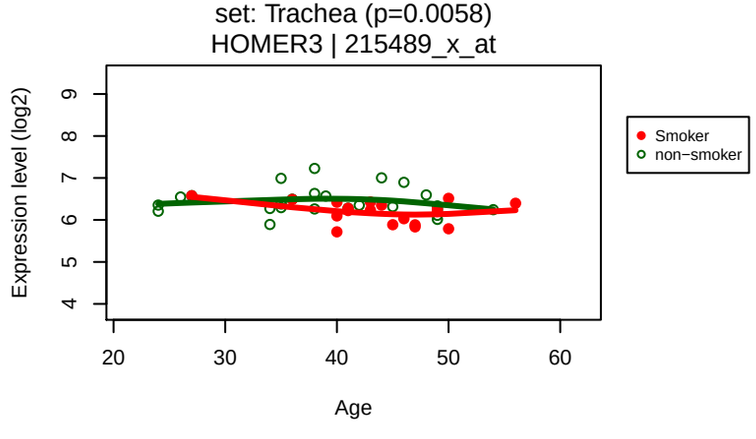

Gene expression ratio in Males
gene: HOMER3 | 215489_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8243
p=0.1055
p=0.0022
p=0.0314
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HOMER3 | 215489_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3936
p=0.9192
p=0.9481
p=0.7832
SmallAirway2
LargeAirway2
Trachea
Macrophage2

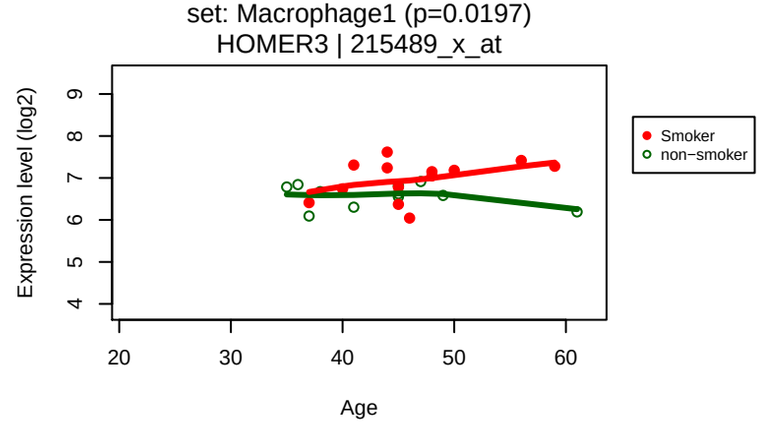

set: SmallAirway1 (p=0.2036)
HOMER3 | 215489_x_at
Expression level (log2)
Age
Smoker
non−smoker

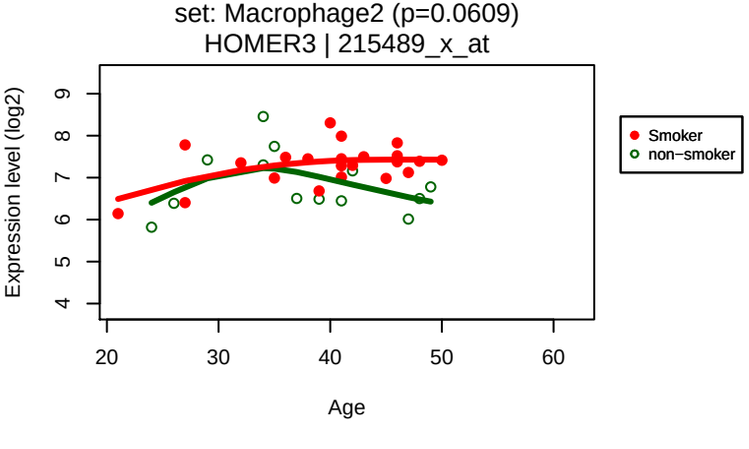

set: SmallAirway2 (p=0.3636)
HOMER3 | 215489_x_at
Expression level (log2)
Age
Smoker
non−smoker

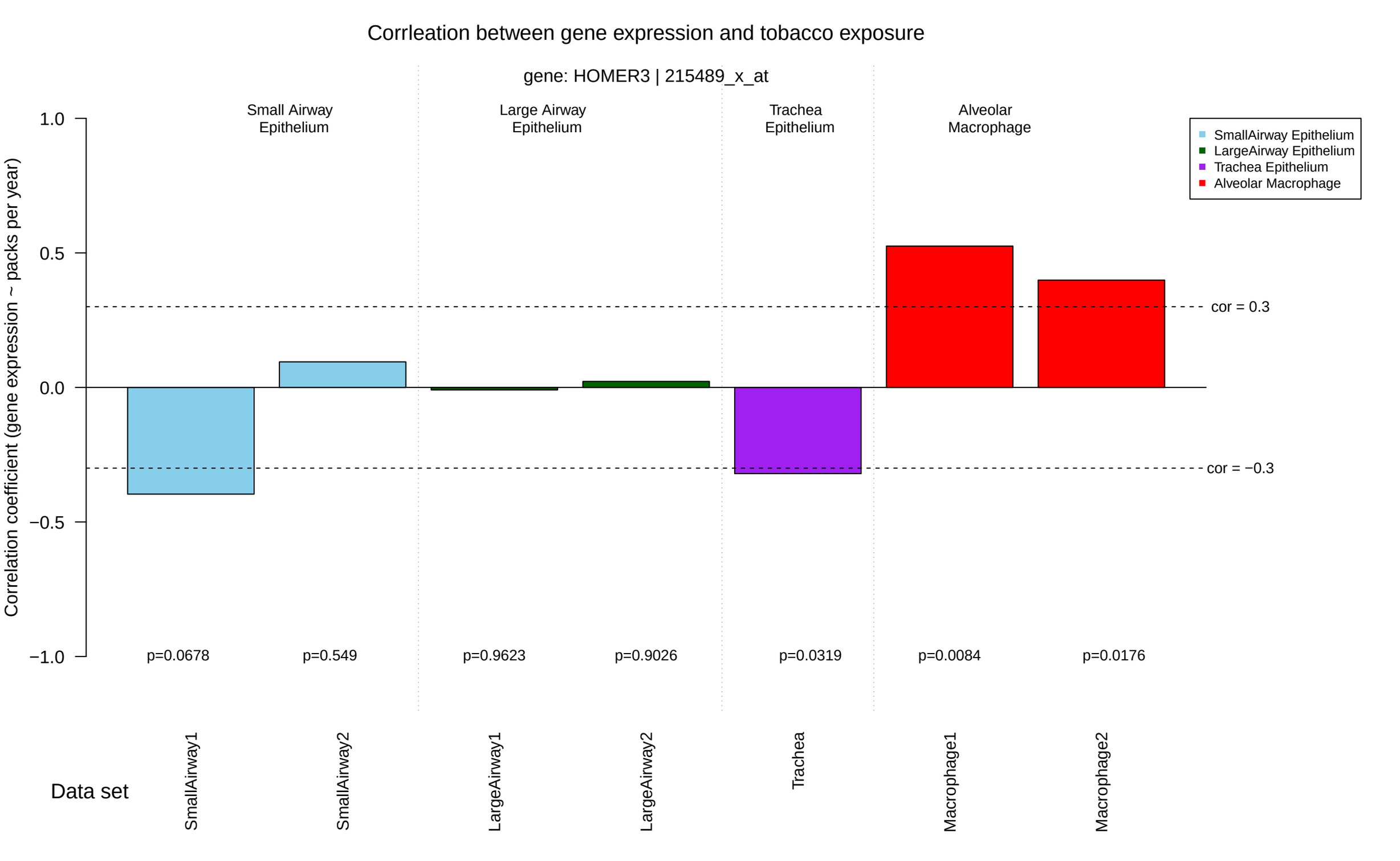

set: LargeAirway1 (p=0.639)
HOMER3 | 215489_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2534)
HOMER3 | 215489_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0058)
HOMER3 | 215489_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0197)
HOMER3 | 215489_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0609)
HOMER3 | 215489_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HOMER3 | 215489_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.0678
p=0.549
p=0.9623
p=0.9026
p=0.0319
p=0.0084
p=0.0176
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2